

Biomedical Morphological Analysis

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH, powersh...
sensitivity sensit
reactive reactiv
measured measur
clinically clinic
relevant relev
intervention intervent
compared compar
physical physic
difference differ
arms arm
respectively respect
differences differ
remained remain
significant signific
rheumatology rheumatolog

Biomedical Words
-----
infection -> infect
infectious -> infecti
infected -> infect
infect -> infect
infecting -> infect
infections -> infect

Result saved successfully.
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Finite-State Morphological Parser

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAME
SH/OneDrive/Documents/booking/q2.py
Downloading Sentiment140...
Download completed.

Words found in dataset:
{'running', 'unreadable', 'unbelievable'}

FST Results
=====
happiest -> Prefix: [] Root: happ Suffix: ['iest']
unbelievable -> Prefix: ['un'] Root: believ Suffix: ['able']
running -> Prefix: [] Root: runn Suffix: ['ing']
reordering -> Prefix: ['re'] Root: order Suffix: ['ing']
smartphones -> Prefix: [] Root: smartphone Suffix: ['s']
unreadable -> Prefix: ['un', 're'] Root: ad Suffix: ['able']

Accuracy
Before correction: 16.666666666666664 %
After correction : 83.33333333333334 %
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Stemming in News Article Classification

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH, powersh...
uments/booking/q3.py
be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://huggingface.co/docs/hugg
ow-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate dev
see this article: https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development
warnings.warn(message)
data/train-00000-of-00001.parquet: downloading bytes: | 18.5MB
data/train-00000-of-00001.parquet: reconstructing file: 100%| 18.6MB / 18.6MB
data/test-00000-of-00001.parquet: downloading bytes: | 1.23MB
data/test-00000-of-00001.parquet: reconstructing file: 100%| 1.23MB / 1.23MB
Generating train split: 100%| 120000/120000 [00:00<00:00, 571616.02
Generating test split: 100%| 7600/7600 [00:00<00:00, 248778.30

RESULTS
-----
Without Stemming : Vocabulary = 13198 Accuracy = 0.8835
Porter Stemming : Vocabulary = 9033 Accuracy = 0.8855
Lemmatization : Vocabulary = 11347 Accuracy = 0.8895

Stemming Examples
-----
organization -> organ
organizer -> organ
organizing -> organ
organized -> organ
organization -> organ
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Morphological Error Analysis in Information Retrieval

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/q4.py
warnings.warn(message)
test.jsonl: downloading bytes: | 121MB, 368kB/s
test.jsonl: reconstructing file: 100%| 195MB / 195MB, 4.82MB/s
Generating train split: 100%| 3600000/3600000 [00:22<00:00, 159893.33 examples/s]
Generating test split: 100%| 400000/400000 [00:02<00:00, 146437.48 examples/s]
Dataset loaded.

Words found:
{'washer', 'watching', 'washable', 'watches', 'washed'}

Stemming Error Analysis
=====
watches -> watch -> Inflectional
watching -> watch -> Inflectional
washable -> washabl -> Derivational
washer -> washer -> Derivational
washed -> wash -> Inflectional

Improved Strategy
=====
watches -> watch
watching -> watching
washable -> washable
washer -> washer
washed -> washed
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Error Analysis of Porter Stemming

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/q5.py
20 STEMMING CASES
=====
organization -> organ
organizer -> organ
organizing -> organ
organized -> organ
organizations -> organ
organizational -> organiz
running -> run
runner -> runner
studies -> studi
studied -> studi
studying -> studi
connection -> connect
connected -> connect
connecting -> connect
easily -> easilli
fairness -> fair
happiness -> happi
management -> manag
manager -> manag
development -> develop

Processed file saved:
C:\Users\K.RAMESH\OneDrive\Documents\booking\Q1_PubMed_Stemming_Processed.csv
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Error Analysis of Finite-State Morphological Parser

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/q6.py
man -> Root: man Type: Singular
men -> Root: man Type: Plural
woman -> Root: woman Type: Singular
women -> Root: woman Type: Plural
mouse -> Root: mouse Type: Singular
mice -> Root: mouse Type: Plural

WORDNET VERIFICATION
=====
car -> Found in WordNet
cars -> Found in WordNet
box -> Found in WordNet
boxes -> Found in WordNet
city -> Found in WordNet
cities -> Found in WordNet
dog -> Found in WordNet
dogs -> Found in WordNet
child -> Found in WordNet
children -> Found in WordNet
man -> Found in WordNet
men -> Found in WordNet
woman -> Found in WordNet
women -> Found in WordNet
mouse -> Found in WordNet
mice -> Found in WordNet
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```

Morphological Feature Extraction

```
PS C:\Users\K.RAMESH\OneDrive\Documents\booking> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe c:/Users/K.RAMESH/OneDrive/Documents/booking/q7.py
=====
QUESTION 7 RESULTS
=====

BEFORE CORRECTION
Vocabulary Size : 46
Accuracy : 100.0 %
Processing Time : 0.0947 seconds

AFTER CORRECTION
Vocabulary Size : 40
Accuracy : 100.0 %
Processing Time : 0.0107 seconds

STEMMING EXAMPLES
=====
running -> run
runners -> runner
runs -> run
studies -> studi
studied -> studi
studying -> studi
organization -> organ
organized -> organ
organizer -> organ
PS C:\Users\K.RAMESH\OneDrive\Documents\booking>
```